

HOMEWORK 5

INTRODUCTION TO COMPLEX ANALYSIS

(MATH 4230-40), SPRING 2022

SECTION 2.1 - CONTOUR INTEGRALS

1. Evaluate the following:

(a) $\int_{\gamma} y dz$, where γ is the union of the line segments joining 0 to i and then i to $i + 2$.

Note that the line segment between 0 and i can be parameterized as $\gamma_1(t) = (1-t)(0) + it = it$ for $\gamma_1 : [0, 1] \rightarrow \mathbb{C}$ and that the line segment between i and $i + 2$ as $\gamma_2(t) = (1-t)i + t(i + 2)$ for $\gamma_2 : [0, 1] \rightarrow \mathbb{C}$. As such, we have

$$\begin{aligned} \int_{\gamma} y dz &= \int_0^1 \operatorname{Im}(\gamma_1(t)) \gamma_1'(t) dt + \int_0^1 \operatorname{Im}(\gamma_2(t)) \gamma_2'(t) dt \\ &= \int_0^1 (t)(i) dt + \int_0^1 (1)(2) dt \\ &= i \int_0^1 t dt + \int_0^1 2 dt \\ &= \frac{i}{2} + 2. \end{aligned}$$

(b) $\int_{\gamma} \sin 2z dz$, where γ is the line segment joining $i + 1$ to $-i$.

Note that the line segment between $i + 1$ and $-i$ can be parameterized as $\gamma(t) = (1-t)(i + 1) + t(-i) = i + 1 - t(2i + 1)$ for $\gamma : [0, 1] \rightarrow \mathbb{C}$. As such, we have

$$\begin{aligned} \int_{\gamma} \sin 2z dz &= \int_0^1 \sin(2\gamma(t)) \gamma'(t) dt \\ &= \int_{2\gamma(0)}^{2\gamma(1)} \frac{1}{2} \sin(u) du \quad (\text{Substituting } u = 2\gamma(t), \text{ so that } 1/2 du = \gamma'(t) dt) \\ &= -\frac{1}{2} \cos(-2i) + \frac{1}{2} \cos(2i + 1). \end{aligned}$$

(c) $\int_{\gamma} ze^{z^2} dz$, where γ is the unit circle.

Note that the functions $f_1(z) = z$, $f_2(z) = e^z$, and $f_3(z) = z^2$ are all entire. Hence, the function $ze^{z^2} = f_1(z)f_2(f_3(z))$ is also entire, and the version of Cauchy's Theorem provided in Theorem 2.2.1 yields that we have $\int_C ze^{z^2} dz = 0$ for any closed curve C . Since γ is such a closed curve, it then follows that $\int_{\gamma} ze^{z^2} dz = 0$

2. Is it true that $\operatorname{Re}\left(\int_{\gamma} f dz\right) = \int_{\gamma} \operatorname{Re} f dz$?

No, it is not true that $\operatorname{Re}\left(\int_{\gamma} f dz\right) = \int_{\gamma} \operatorname{Re} f dz$. There are many counterexamples—for an easy one, consider the case where γ is the straight line from 0 to i parameterized as $\gamma(t) = it$ for $t \in [0, 1]$ and $f(z) = \operatorname{Im}(z)$. Then,

$$\begin{aligned}\operatorname{Re}\left(\int_{\gamma} f dz\right) &= \operatorname{Re}\left(\int_0^1 f(\gamma(t))\gamma'(t)dt\right) \\ &= \operatorname{Re}\left(\int_0^1 (t)(i)dt\right) \\ &= \operatorname{Re}\left(\frac{i}{2}\right) \\ &= 0.\end{aligned}$$

However, since $\operatorname{Re}(\operatorname{Im}(z)) = \operatorname{Im}(z)$ (since $\operatorname{Im}(x + iy) = y$), we have

$$\begin{aligned}\int_{\gamma} \operatorname{Re} f dz &= \int_0^1 \operatorname{Re}(f(\gamma(t)))\gamma'(t)dt \\ &= \int_0^1 \operatorname{Re}(\operatorname{Im}(\gamma(t)))\gamma'(t)dt \\ &= \int_0^1 (t)(i)dt \\ &= \frac{i}{2}.\end{aligned}$$

3. Evaluate $\int_{\gamma} x^2 - y^2 dz$, where γ is the straight line from 0 to i .

Note that the line segment between 0 and i can be parameterized as $\gamma(t) = (1-t)(0) + t(i) = ti$. As such, we have

$$\begin{aligned}\int_{\gamma} x^2 - y^2 dz &= \int_0^1 ((0)^2 - (t)^2)\left(\frac{d}{dt}ti\right)dt \\ &= \int_0^1 (-t^2)(i)dt \\ &= -i \int_0^1 t^2 dt \\ &= -\frac{i}{3}\end{aligned}$$

4. Let γ be the arc of the circle $|z| = 2$ that lies in the first quadrant. Show that

$$\left| \int_{\gamma} \frac{1}{z^2 + 1} dz \right| \leq \frac{\pi}{3}.$$

We use proposition 2.1.6 to show that if $|f(z)| \leq M$ for z on the curve γ , then $\left| \int_{\gamma} f(z) dz \right| \leq M\ell(\gamma)$, where $\ell(\gamma)$ denotes the length of γ . To do so, note that the circle $|z| = 2$ has circumference 4π , so if γ is the arc that lies in the first quadrant, then its length is a quarter of this value; i.e., $\ell(\gamma) = \pi$. Similarly, note that for $z = 2e^{it}$, we have

$$\begin{aligned} |z^2 + 1| &= |(\operatorname{Re} z)^2 + 2i\operatorname{Re} z \operatorname{Im} z + 1 - (\operatorname{Im} z)^2| \\ &\geq \left| |(\operatorname{Re} z)^2 + 2i\operatorname{Re} z \operatorname{Im} z + 1| - |(\operatorname{Im} z)^2| \right| \quad (\text{Reverse Triangle Inequality}) \\ &\geq \left| |1| - |2^2| \right| \quad (\text{Since } 0 \leq \operatorname{Re} z \leq 2 \text{ and } 0 \leq \operatorname{Im} z \leq 2.) \\ &= |1 - 4| \\ &= 3. \end{aligned}$$

This yields that $\frac{1}{|z^2 + 1|} = \left| \frac{1}{z^2 + 1} \right| \leq M = \frac{1}{3}$ on the first quadrant of the circle $|z| = 2$. Hence, by Proposition 2.1.6, we have

$$\left| \int_{\gamma} \frac{1}{z^2 + 1} dz \right| \leq M\ell(\gamma) = \frac{\pi}{3}.$$

SECTION 2.2 - CAUCHY'S THEOREM - A FIRST LOOK

5. For what simple closed curves γ does the equation $\int_{\gamma} \frac{dz}{z^2 + z + 1} = 0$ hold?

Note that $f(z) = \frac{1}{z^2 + z + 1}$ is analytic on $\mathbb{C} \setminus \left\{ -\frac{1}{2} \pm \frac{\sqrt{3}}{2}i \right\}$. As such, for any simple closed curve γ which does not enclose or pass through either $z_1 = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$ or $z_2 = -\frac{1}{2} - \frac{\sqrt{3}}{2}i$, the version of Cauchy's Theorem provided in Theorem 2.2.1 yields that $\int_{\gamma} \frac{dz}{z^2 + z + 1} = 0$. As such, we need only ascertain the value of $\int_{\gamma} \frac{dz}{z^2 + z + 1}$ for a simple closed curve γ bounding a region which contains either z_1 , z_2 , or both z_1 and z_2 . By The Deformation Theorem, it will suffice to take γ as a circle in each case. Noting that $|z_1| = |z_2| = 1$ and $|z_1 - z_2| = |i\sqrt{3}| \approx 1.73$, we will use a circle of radius 1 centered at z_1 , a circle of radius 1 centered at z_2 , and a circle of radius 2 centered at the origin for this purpose.

Proceeding with the case of z_1 , we let $\gamma_1(t) = z_1 + e^{2\pi it}$ for $t \in [0, 1]$ and compute

$$\begin{aligned} \int_{\gamma_1} \frac{dz}{z^2 + z + 1} dz &= \int_{\gamma_1} \frac{dz}{(z - z_1)(z - z_2)} \\ &= \int_0^1 \frac{1}{(e^{2\pi it})(e^{2\pi it} + z_1 - z_2)} 2\pi i e^{2\pi it} dt \\ &= \int_0^1 \frac{2\pi i}{e^{2\pi it} + i\sqrt{3}} dt \\ &= \int_0^1 \frac{2\pi i}{e^{2\pi it} + i\sqrt{3}} dt \\ &= \frac{2\pi}{\sqrt{3}} \neq 0. \end{aligned}$$

For the case of z_2 , we let $\gamma_2(t) = z_2 + e^{2\pi it}$ for $t \in [0, 1]$ and compute

$$\begin{aligned} \int_{\gamma_2} \frac{dz}{z^2 + z + 1} dz &= \int_{\gamma_2} \frac{dz}{(z - z_1)(z - z_2)} \\ &= \int_0^1 \frac{1}{(e^{2\pi it} + z_2 - z_1)(e^{2\pi it})} 2\pi i e^{2\pi it} dt \\ &= \int_0^1 \frac{2\pi i}{e^{2\pi it} - i\sqrt{3}} dt \\ &= -\frac{2\pi}{\sqrt{3}} \neq 0. \end{aligned}$$

Finally, in the case of both z_1 and z_2 , we let $\gamma_3(t) = 2e^{2\pi it}$ for $t \in [0, 1]$ and compute

$$\begin{aligned} \int_{\gamma_3} \frac{dz}{z^2 + z + 1} dz &= \int_0^1 \frac{1}{4e^{4\pi it} + 2e^{2\pi it} + 1} 4\pi i e^{2\pi it} dt \\ &= \int_0^1 \frac{1}{4e^{4\pi it} + 2e^{2\pi it} + 1} 4\pi i e^{2\pi it} dt \\ &= 0. \end{aligned}$$

6. Let γ_1 be the circle of radius 1 and let γ_2 be the circle of radius 2 (traversed counterclockwise and centered at the origin). Show that

$$\int_{\gamma_1} \frac{dz}{z^3(z^2+10)} = \int_{\gamma_2} \frac{dz}{z^3(z^2+10)}.$$

Note that $f(z) = \frac{1}{z^2(z^2+10)}$ is analytic on the set $\mathbb{C} \setminus \{0, \sqrt{10}i, -\sqrt{10}i\}$. Note as well that both γ_1 and γ_2 are simple closed curves that can be parameterized by $\gamma_1(\theta) = e^{2\pi i\theta}$ and $\gamma_2(\theta) = 2e^{2\pi i\theta}$, where $\theta \in [0, 1]$. The function $h(t, \theta) = (1+t)e^{2\pi i\theta}$ is then a homotopy between γ_1 and γ_2 , as h is continuous and satisfies both $h(0, \theta) = e^{2\pi i\theta} = \gamma_1(\theta)$ and $h(1, \theta) = 2e^{2\pi i\theta} = \gamma_2(\theta)$. Since $h(t, \theta) \in A$ for any $(t, \theta) \in [0, 1] \times [0, 1]$ (this can be seen by noting that $1 \leq |h(t, \theta)| \leq 2$ and that $|\pm \sqrt{10}i| > 2$), the Deformation Theorem (i.e. Theorem 2.2.2) yields that

$$\int_{\gamma_1} \frac{dz}{z^3(z^2+10)} = \int_{\gamma_2} \frac{dz}{z^3(z^2+10)}.$$

7. Evaluate $\int_{\gamma} \frac{2z^2 - 15z + 30}{z^3 - 10z^2 + 32z - 32} dz$ where γ is the circle $|z| = 3$. Hint: Use partial fractions; one root of the denominator is $z = 2$.

Using partial fractions, we produce $\frac{2z^2 - 15z + 30}{z^3 - 10z^2 + 32z - 32} = \frac{2}{z-2} + \frac{1}{(z-4)^2}$. Accordingly, parameterizing γ as $\gamma(t) = 3e^{2\pi it}$ for $t \in [0, 1]$, we compute

$$\begin{aligned} \int_{\gamma} \frac{2z^2 - 15z + 30}{z^3 - 10z^2 + 32z - 32} dz &= \int_{\gamma} \frac{2}{z-2} + \frac{1}{(z-4)^2} dz \\ &= \int_{\gamma} \frac{2}{z-2} dz + \int_{\gamma} \frac{1}{(z-4)^2} dz \\ &= \int_{\gamma} \frac{2}{z-2} dz \quad \left(\text{Since } \frac{1}{(z-4)^2} \text{ is analytic on and inside } \gamma. \right) \\ &= \int_0^1 \frac{2(6\pi i e^{2\pi it})}{3e^{2\pi it} - 2} dt. \end{aligned}$$

By the Deformation Theorem, we may replace γ with the circle $\tilde{\gamma}$ centered at 2 of radius 1. Accordingly, a parameterization for $\tilde{\gamma}$ is $\tilde{\gamma}(t) = 2 + e^{2\pi it}$ and we may compute

$$\begin{aligned} \int_{\gamma} \frac{2}{z-2} dz &= \int_{\tilde{\gamma}} \frac{2}{z-2} dz \\ &= \int_0^1 \frac{2(2\pi i e^{2\pi it})}{2 + e^{2\pi it} - 2} dt \\ &= \int_0^1 4\pi i dt \\ &= 4\pi i. \end{aligned}$$